MYP 3 - Mathematics

Practice Sheet - Unit 1: Laws of Exponents (Criterion A)

Name: _____ Date: _____

Instructions: Show all your working. Only a simple calculator (4 operations) is allowed. Unless a question says otherwise, write all final answers using **positive exponents only**.

Level 1-2 Solve simple problems in familiar situations.

1. Match each law of exponents on the left with its name on the right. Write the letter in the answer row.

Law	Name
1. $a^m \times a^n = a^{m+n}$	A. Quotient law
2. $\frac{a^m}{a^n} = a^{m-n}$	B. Zero exponent law
3. $(a^m)^n = a^{m \times n}$	C. Product law
4. $a^{-n} = \frac{1}{a^n}$	D. Negative exponent law
5. $a^0 = 1,$ where $a \neq 0$	E. Power of a power law

Answers: 1 → ____ 2 → ____ 3 → ____ 4 → ____ 5 → ____

2. Find the value of each of the following. Give your answers as whole numbers or fractions.

(a) 5^{-3}	(b) 9^0

3. Select the correct simplification for each expression by ticking one box.

(a) Simplify	$x^4 \times x^3$	[ ]	x^{12}	[ ]	x^7	[ ]	x^1
(b) Simplify	$\frac{y^6}{y^2}$	[ ]	y^3	[ ]	y^4	[ ]	y^8
(c) Simplify	$(z^2)^5$	[ ]	z^{10}	[ ]	z^7	[ ]	z^{25}

Level 3-4	Solve more complex problems in familiar situations.

4. Find the value of each of the following. Give your answers as whole numbers or fractions.

(a) -3^{-3}	(b) $\left(\frac{1}{3}\right)^{-3}$
(c) $\left(\frac{6}{7}\right)^{-2}$	(d) $-9^0 + (-9)^0$

5. Simplify each expression. Write your answers using positive exponents only.

(a) $3^{-2}x^{-2}y^3z^{-11}t^0$	(b) $(2x^3)^4$
(c) $4r^{-3} \times 2r^2$	(d) $(2x^4y^{-3})^{-1}$

Level 5-6 Solve challenging problems in familiar situations.

6. Simplify the following expression, writing your answer with positive exponents only:

$$\frac{3x^{-2}y^{3}}{6xy^{-4}}$$

7. Simplify fully, writing your answer with positive exponents only:

$$\frac{x^{2}y^{-1}}{z^{3}} \times \frac{(xz)^{2}}{y^{4}}$$

8. Simplify fully, writing your answer with positive exponents only:

$$\frac{m^{3}n^{2}p}{p^{4}q} \div \frac{m^{5}n^{2}}{(pq)^{3}}$$

9. A student simplified the expression $\frac{(2a^3)^2}{4a^{-2}}$ as follows:

Step 1: $\frac{(2a^3)^2}{4a^{-2}} = \frac{2a^6}{4a^{-2}}$

Step 2: $\frac{2a^6}{4a^{-2}} = \frac{a^{6-2}}{2} = \frac{a^4}{2}$

(a) **Identify** the mistake in Step 1 and the mistake in Step 2, and **explain** each one.
(b) **Find** the correct simplified answer. Show all your working.

Level 7-8	Solve challenging problems in both familiar and unfamiliar situations.

10. A large cube has an edge length of $4x^2y$ cm.

Assume that x and y are positive integers.

(a) **Find** the volume of the large cube, in terms of *x* and *y*.

(b) The large cube is cut into identical small cubes, each with an edge length of $2xy^{-1}$ cm.

Find the number of small cubes, giving your answer with positive exponents only.

End of the assessment